

Law of Sines and Law of Cosines

ANSWER KEY



Law of Sines

- 1 In triangle ABC , $A = 40^\circ$, $B = 65^\circ$, and $a = 12$. Find:

a angle $C = 75^\circ$

b side b (2 d.p.)
 $= \frac{12 \sin 65^\circ}{\sin 40^\circ} \approx 16.92$

c side c (2 d.p.)
 $= \frac{12 \sin 75^\circ}{\sin 40^\circ} \approx 18.03$

- 2 Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are 58° and 47° . Find the distance from the balloon to the first observer (the one with the 58° sighting), to the nearest metre.

The angle at the balloon is $180^\circ - 58^\circ - 47^\circ = 75^\circ$. By the Law of Sines, $d = \frac{400 \sin 47^\circ}{\sin 75^\circ} \approx 303$ m.

Law of Cosines

- 3 In triangle PQR , two sides of length 8 and 11 enclose an angle of 52° . Find the length of the third side, to two decimal places.

$$c^2 = 8^2 + 11^2 - 2(8)(11)\cos 52^\circ \approx 185 - 108.36 = 76.64, \text{ so } c \approx 8.75.$$

- 4 A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.

The largest angle is opposite the side of length 12: $\cos C = \frac{7^2 + 9^2 - 12^2}{2(7)(9)} = \frac{-14}{126} = -\frac{1}{9}$, so $C \approx 96.4^\circ$.

- 5 Find the area of a triangle with sides 10 and 14 enclosing an angle of 35° , to two decimal places.

$$A = \frac{1}{2}(10)(14)\sin 35^\circ \approx 70 \times 0.5736 \approx 40.15 \text{ square units.}$$

- 6 A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N 50^\circ E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.

The angle between the legs is $90^\circ + 50^\circ = 140^\circ$. Law of Cosines:

$$d^2 = 30^2 + 45^2 - 2(30)(45)\cos 140^\circ \approx 2925 + 2068.4 = 4993.4, \text{ so } d \approx 71 \text{ km.}$$